CivicProof GIS: Technical Architecture & Design Specification

Systematic Design of Node/SQLite Backend, OSRM Routing & Offline Sync

1. Architectural Overview

CivicProof GIS is engineered as a resilient, full-stack hybrid Progressive Web Application (PWA). It decouples field client operations from continuous server connectivity by implementing a three-tiered state model: In-Memory Zustand Store -> IndexedDB Client Persistence -> Centralized Express/SQLite API.

2. Component Architecture

• Frontend Engine: React 18 with Vite 8 bundling, Tailwind CSS v4, and Leaflet GIS mapping. All components are responsive and optimized for mobile screens (portrait 360px to 430px).
• Routing Subsystem: Integrates with the Open Source Routing Machine (OSRM). Computes real road-network polyline geometries for walking, cycling, and driving modes.
• Apple Maps Navigation UX: Features a high-contrast dark HUD (#1c1c1e), prominent green #34c759 "Start" button, animated pulsing blue navigation puck, step-by-step turn guidance, and Web Speech Synthesis vocalization in Telugu, Hindi, and English.
• Backend Service: Node.js 24 environment utilizing the high-performance native DatabaseSync (node:sqlite) module in WAL mode. Eliminates external C++ compilation dependencies while delivering ACID transactional integrity.
• QR Security & Token Architecture: QR labels utilize cryptographically random nanoid tokens rather than sequential or internal database primary keys. Public endpoints expose only operational data, protecting administrative metadata.

3. Data Flow & Offline Resiliency Pipeline

1. Initial Handshake: Service Worker precaches all core assets (JS, CSS, fonts, SVG icons). Facilities and route queries are stored into IndexedDB via the 'idb' wrapper.
2. Disconnected Operation: When network connectivity drops, the app switches to IndexedDB fallback mode. Facilities show an amber 'Cached Data' banner with last-sync timestamp.
3. Offline Field Inspection Drafts: Inspectors can fill inspection forms and defect items in the field without internet. Drafts are assigned local UUIDs and saved to IndexedDB.
4. Event-Driven Auto Sync: The client listens for the browser 'online' event. As soon as cellular connection is restored, pending drafts are automatically posted in sequence to /api/inspections and marked as synced.

4. Security & Role-Based Access Control (RBAC)

• Roles: Administrator (full oversight, QR generation, facility verification), Inspector (inspections, defect logging, attention queue), Field Staff (inspection drafts).
• Authentication: JSON Web Tokens (JWT) signed with HMAC-SHA256, salted bcrypt password hashing (10 rounds), and HTTP Bearer authorization headers with auto-logout interceptors.